

NIDHIL NAYUDU

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EDUCATION

University of Michigan: <i>5x Dean's List, 6x University Honors</i> Computer Science B.S.E, Minor in Business; GPA: 3.84 Relevant Coursework: Data Structures & Algorithms, Web Systems, Distributed Systems, Operating Systems, Computer Security, Networking, Machine Learning, Computer Science Theory	Aug. 2022 – Dec. 2025 Ann Arbor, MI
University of Michigan Computer Science M.S.E: <i>1-year accelerated Master's program continuing immediately after B.S.E (Expected).</i>	Jan. 2026 – Dec. 2026 Ann Arbor, MI

EXPERIENCE

CGI Software Development Intern - Integrated Model Context Protocol (MCP) into our Agentic AI framework for SDLC automation, decoupling agent logic from client systems to enable server-side updates that cut client integration time by 42%. - Engineered GitLab CI/CD pipelines and core Agentic AI agents (BDD test generation/execution, Dev Environment, Pipeline Orchestration), automating ATDD validation and boosting test coverage by 8% across over 10 modules. - Spearheaded architectural redesign over three Agile sprints, aligning client and executive goals by leading stakeholder meetings and creating system diagrams that halved design approval time and reduced misalignment.	June 2025 - Present New York, NY
Human Corporation Software Engineer Intern - Collaborated with a Ugandan client to streamline sprint planning and redesign workflows, reducing the development cycle time by 15% for a charitable e-commerce platform. - Integrated Firestore authentication and Firestore database to manage over 50 user, product, and order records in production. - Refactored over 10 API endpoints into modular Node.js controllers, improving database query efficiency at scale.	May 2024 – Aug. 2024 New York, NY
University of Michigan EECS 280: Programming & Intro Data Structures Instructional Aide (TA) - Exam Lead - Led lab sections and held office hours to assist over 1,000 students, reinforcing core CS concepts and providing personalized support. - Partnered with faculty to co-develop exams and initiated changes to clarify ambiguous questions, improving student comprehension and grading consistency. - Facilitated understanding of concepts including recursion and polymorphism by refining lecture material and providing TA guidance.	Aug. 2024 - Present Ann Arbor, MI

PROJECTS

JobAlign: Resume Optimizer - Project Lead <i>Python, Machine Learning, NLP, RAG, PostgreSQL</i> - Engineered a custom SpaCy NLP pipeline to parse, match, and heuristically optimize resumes for job postings, using a combination of semantic similarity and TF-IDF scoring, achieving 75% parsing accuracy. - Developed and deployed a Streamlit-based SaaS platform for dynamic job matching, using PostgreSQL as a vector database to store embeddings and perform top-<i>k</i> semantic search. - Implemented a Retrieval-Augmented Generation (RAG) system that retrieves the top 10 most relevant resumes based on user job input, and integrates them as context for GPT-4o to evaluate resume fit and recommend improvements. - Integrated a chatbot offering users personalized feedback and interactive guidance for efficient recruitment strategies.
ask485?: Search Engine <i>Python, React.js, Hadoop, Distributed Systems, Flask, Multithreading, APIs</i> - Created a Google-style scalable search engine with a React.js frontend and Python backend to return ranked web search results. - Built a segmented inverted index using a MapReduce pipeline with Hadoop streaming for efficient query retrieval. - Employed Flask Python framework to develop REST APIs and route requests with backend index and search servers.
Thread Library <i>C++, Multithreading, Unix, OS, Unit Testing</i> - Crafted an advanced C++ thread library with meticulous implementation of Mutexes, Threads, and Conditional Variables, featuring essential functions like lock, unlock, signal, broadcast, wait, thread yield, and thread join. - Implemented memory management mechanisms to prevent leaks and enhance library reliability in concurrent environments. - Enforced comprehensive unit tests to assure the library's soundness, identify and rectify errors, and guarantee optimal performance in diverse scenarios.
AFK DJ (Dance Detection) <i>Python, Computer Vision, Machine Learning, APIs</i> - Built a real-time classifier with CNNs using OpenCV and Mediapipe to analyze landmarked body movements and predict "dancing" or "not dancing" with 90% accuracy. - Established a 10 second buffer to pause Spotify playback if dancing dropped, improving audience responsiveness to song quality. - Integrated Spotify API to automate playlist control based on detected engagement levels.

TECHNICAL SKILLS

Languages	: Python, C/C++, JavaScript/TypeScript, HTML/CSS, SQL
Technologies	: Next.js/React.js, Git, gdb, Linux, Flask, Multithreading, Networks, AWS, Tensorflow, PyTorch, Docker
Concepts	: Object-Oriented Programming, Software Dev, Machine Learning, AI, REST APIs, Agile Dev
Interests	: Journaling, Self-learning Telugu, Finance Podcasts, Basketball/Gym, River Tubing